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Lab 2 Report

- Observed that both light and proximity sensors are arranged in a circular configuration around the robot.
- Initially grouped sensors manually into front, left, right, and back categories using hard-coded arrays only to realize that hard-coding sensor groups was not scalable or elegant. An angle-based mathematical representation was used in the end.
- Expected the angle-based solution to provide the same functionality while being cleaner and easier to maintain.
- Designed the controller to:
    - Identify the strongest light sensor.
    - Identify the strongest proximity sensor.
    - Prioritize obstacle avoidance whenever proximity exceeds a threshold.
    - Follow the light source when no obstacle is detected.
    - Perform random exploration when no significant light is detected.
    - Implemented velocity smoothing using a weighted average between previous and desired wheel velocities.
- Expected smoothing to reduce oscillations and improve movement stability.
- Hypothesized that the robot would successfully reach the light while avoiding collisions.
- Compared behavior with and without sensor/actuator noise.
- Expected noise to reduce performance.
- Observed that moderate noise sometimes improved behavior.
- Noise prevented the robot from getting trapped in repetitive movement patterns.
- Small random variations helped the robot escape local behaviors and continue toward the target.
- Tested different light intensity values.
- Observed that high light intensity strongly attracted the robot toward the target.
- In some cases, obstacle avoidance appeared less important because the light dominated decision-making.
- Lower light intensities produced a more balanced behavior between phototaxis and obstacle avoidance.